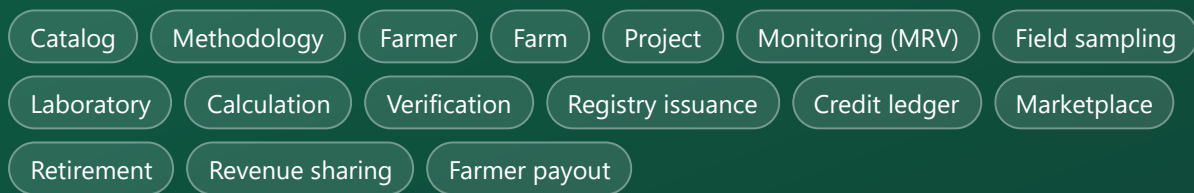


Carbon dMRV Platform

End-to-End Product and Test Guide

How the platform turns farm-level soil and land-management data into verified, issued and sold carbon credits, and pays farmers their share, with each stage explained and its tests listed.



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About this document

The document covers three things:

- **The product.** What the platform does at each stage of the carbon-credit lifecycle, and who does it.
- **The controls.** The approval, audit and data-integrity rules that make the output defensible to a verifier, a registry and a buyer.
- **The test procedure.** How a tester walks the full lifecycle in the application, what they should see at every step, and which actions must be *refused*.

The test results in Chapter 8 come from complete runs of the lifecycle on the local test environment on 4–5 October 2026, using dedicated test accounts.

Important status note

All carbon calculation modules are labelled **Not production ready** until a methodology expert has reviewed each one against the official methodology document and the platform has recorded that sign-off. The test credits in this document were issued against a test registry organization and are not real credits. See Chapter 9.

1. Executive summary

The Carbon dMRV Platform is a digital **measurement, reporting and verification (dMRV)** system for nature-based carbon projects, focused on agriculture and smallholder farmers. It covers the whole lifecycle in one application:

- registering farmers and their land;
- measuring soil carbon and farm practices in the field and in the laboratory;
- calculating emission reductions and removals under recognised methodologies;
- verification by an independent body;
- recording registry issuance;
- selling and retiring the credits;
- paying farmers their share of the revenue.

17

lifecycle stages, from catalog
to farmer payout

13

calculation modules across
Verra, Gold Standard and
India CCTS

24

distinct business roles with
enforced separation of
duties

705

automated tests passing
(552 backend + 153
frontend)

What was demonstrated

- **Complete lifecycle, end to end.** A VM0042 soil-carbon project went through every stage:
 - farmer and farm onboarding;
 - two soil-sampling campaigns with control sites, and laboratory analysis;
 - a calculation of **24 VCU**;
 - independent verification of 24 tCO₂e;
 - registry issuance of 24 VCU, confirmed by a second person, and entry into the credit ledger;
 - a sale of 10 VCU to a buyer for INR 8,500, of which the buyer retired 4 VCU;
 - a revenue settlement, with a **farmer payout of INR 4,200**, paid and reconciled against the bank statement and visible to the farmer in their own login.
- **Every methodology option calculates.** All 13 methodology options (one calculation module each) were run end to end. These include Gold Standard soil carbon (five variants), tree planting on farms, biochar, India CCTS afforestation and mangroves, rice methane and Gold Standard afforestation.
- **Separation of duties holds.** Every negative test passed: the person who submits a record can never approve it. This covers methodology approval, farmer KYC, farm verification, eligibility, field records, lab results, calculations, issuance, ledger openings, buyer payments and settlements.

What remains before production use

- Expert sign-off of each calculation module against the official methodology (the platform workflow for this exists; the reviews have not been done).
- Gold Standard Paris-alignment status for vintages from 2026 onwards. The platform enforces this gate today.
- A decision on splitting one farm into several strata (research is complete; the decision is pending).
- Production infrastructure items listed in Chapter 9: malware-scanning engine, multi-factor sign-in, shared rate-limit store and live payment and registry integrations.

2. What the platform does

2.1 The problem it solves

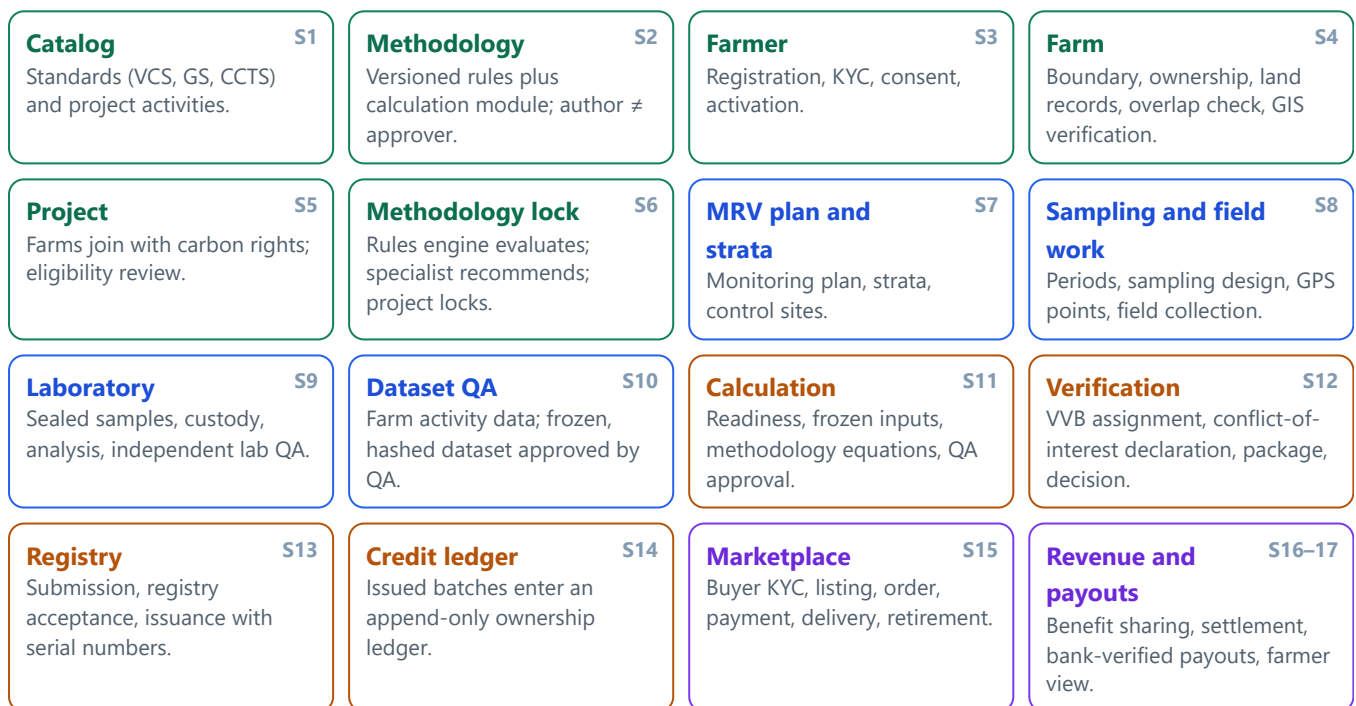
Carbon credits from agriculture depend on proving, field by field, that a change in practice stored more carbon in the soil or cut emissions. The proof must survive three reviewers:

- an independent **verification body (VVB)**;
- a **registry** (Verra, Gold Standard or India's CCTS);
- the **buyer**.

Each needs to trace every credit back to a sample taken at a known GPS point on a known farm, analysed by a named laboratory, under a fixed version of the rules. Spreadsheets cannot keep that chain intact. The platform keeps every link as a database record that cannot be silently changed afterwards.

2.2 The lifecycle at a glance

Colour by phase: ■ Set-up and onboarding ■ Measurement (MRV) ■ Quantify, verify, issue ■ Commercial and payout



2.3 The evidence chain

At any point a reviewer can follow a credit back to its source. In the application this is the **Lineage** view, available on lab results, calculation runs, verification decisions, issuance batches, orders and payouts:

Farmer payout → settlement run → revenue record → order and payment → ledger transfer → credit batch (vintage, serial range) → registry issuance → registry submission (SHA-256 snapshot) → VVB decision → verification package → calculation report → approved calculation run → frozen inputs → laboratory result → sealed sample → field collection (GPS, photo) → sampling point → stratum → farm → farmer

2.4 Technology in one paragraph

The platform has three layers:

- **Web application:** Angular, served to the browser.
- **Server API:** Python/FastAPI.
- **Database:** Microsoft SQL Server, whose geography engine measures farm areas, detects overlapping boundaries and validates GPS points.

Calculations use exact decimal arithmetic, and every step records which methodology equation produced it. Documents are stored with a SHA-256 checksum, and key tables are protected by database triggers so that history cannot be edited or deleted.

3. Organizations, roles and separation of duties

Every user belongs to an organization and holds one or more roles. A role grants permissions, such as "approve KYC" or "confirm payment". The menu only shows the pages a user's permissions allow, and the server re-checks every request.

3.1 Organization types

Organization	Responsibility in the lifecycle	Test organization
Platform	Runs the platform: administration, catalog, methodology specialists, marketplace compliance.	PLATFORM
Project developer	Onboards farmers and farms, runs projects and monitoring, calculates, submits to verification and the registry, sells credits, pays farmers.	DEV: Green Farms Developer
Farmer group / FPO	Farmer organization; farmers sign in to see their farms, projects and payouts.	E2EFPO
Laboratory	Receives sealed samples, analyses them, performs independent lab QA. Sees no farmer names or GPS locations.	LAB: Soil Test Lab
VVB / ACVA	Independent validation and verification body. Sees only the package it was assigned, through a separate restricted interface.	VVB: Verify Co
Registry	External counterparty (Verra, Gold Standard, CCTS). Its responses are recorded with reference numbers and PDF evidence.	E2EREG: test registry
Buyer	Buys credits on the marketplace after KYC; holds and retires them.	BUY: Buyer Co

3.2 Roles and what each one does

Role	Organization	Main responsibilities
Platform Admin	Platform	Organizations, users, roles, consent types, standards and activities catalog, audit log, background jobs, security center.
Methodology Specialist (×2)	Platform	One writes a methodology version and its rules; a second approves it. Recommends the methodology for a project.
Marketplace Compliance	Platform	Verifies buyer KYC.
Project Manager	Developer	Registers farmers and farms, creates projects, confirms the methodology lock, proposes the VVB, records benefit-sharing terms and costs.
QA Officer	Developer	Approves KYC, carbon rights, eligibility, MRV plan, datasets and calculations; confirms issuances, ledger openings and deliveries.
GIS Specialist	Developer	Verifies farm boundaries and ownership, accepts project boundaries, approves strata and sampling designs.
MRV Manager	Developer	Creates the monitoring plan, periods, strata, sampling designs, records monitoring data, prepares datasets, engages the laboratory.
Field Supervisor	Developer	Assigns sampling points to collectors, accepts field records, ships samples.

Role	Organization	Main responsibilities
Field Collector / Agent	Developer	Collects samples at assigned points on a phone (GPS, photos, checklist), registers and seals samples. Approves nothing.
Calculation Analyst	Developer	Runs calculations: readiness check, input freeze, execution, response to QA findings.
Registry Manager	Developer	Registry accounts, project registration, issuance submissions, records issuance.
Credit Manager	Developer	Opens issued batches in the ledger, creates marketplace listings.
Finance Manager (×4 in test)	Developer	Approves listings, confirms buyer payments, approves costs and sharing terms, calculates and approves settlements, approves and executes payouts, reconciles.
Lab Manager / Lab QA	Laboratory	Accepts engagements; a separate person performs independent QA of results.
Lab Technician	Laboratory	Receives samples, checks seals, registers, analyses, uploads the lab report.
VVB Reviewer (×2)	VVB	Accepts assignments with a conflict-of-interest declaration, reviews the package, raises findings. The decision is recorded by a reviewer who did not raise findings on that package.
Buyer	Buyer	Buyer profile and KYC, orders, payment proof, holdings, retirement requests.
Farmer	Farmer group	Self-service: own profile, farms, project participation and payouts.

3.3 Separation of duties (the "four-eyes" principle)

The platform enforces that **whoever submits or records something can never approve or confirm it**. The server enforces this, not the screen: a missing button is a convenience, and the server would refuse the action anyway with the code `SEPARATION_OF_DUTIES`.

Action	Done by	Must be approved / confirmed by a different person
Methodology version	Author specialist	Second specialist
Calculation module production readiness	Requesting specialist	Second specialist
Farmer KYC	Project manager	QA officer
Farm verification	Submitter	GIS specialist
Project eligibility, carbon rights	Project manager	QA officer
Methodology confirmation	Recommended by a specialist	Confirmed by the project manager (the recommender cannot lock)
MRV plan, stratum, sampling design, dataset	MRV manager	QA officer / GIS specialist
Field record	Collector	Supervisor
Lab engagement / lab result	Proposer / technician	Lab manager / lab QA
Calculation run	Analyst	QA officer
VVB findings and decision	VVB reviewer who raised findings	Second VVB reviewer records the decision
Registry issuance, ledger opening	Registry / credit manager	QA officer
Listing, buyer payment	Credit manager / buyer	Seller finance
Settlement, payout	Finance calculator	A second finance manager approves; a third executes
Farmer bank account	Project manager adds	Finance verifies

4. Controls that apply everywhere

AUDIT TRAIL

Every action is logged with the user, time, organization and the **reason** they typed. Audit tables are append-only, enforced by database triggers. See **Administration** → **Audit log**.

REASONS REQUIRED

Almost every state change asks for a reason of at least 3 characters, which is kept in the history.

"STILL NEEDED" CHECKLISTS

Detail pages show what is missing before the next step is allowed. Errors name the business rule, e.g. `REQUIREMENTS_NOT_MET`, and the field at fault.

STATUS HISTORY

Projects, farms, periods, runs and other records keep a full status history: who moved it, when and why.

FROZEN, HASHED SNAPSHOTS

Datasets, calculation inputs, calculation reports, verification packages and registry submissions are frozen and fingerprinted with SHA-256. Any later change is detected.

IMMUTABLE APPROVED RECORDS

An approved methodology version, an approved lab result or an issued batch is never edited. Changes create a new version that supersedes the old one, and both stay visible.

DOCUMENTS

The file type is detected from the content, not the file name. Every file gets a SHA-256 checksum and a malware-scan hook, and suspicious files go to **Document quarantine**. KYC and bank documents are restricted.

PERSONAL DATA PROTECTION

KYC ID numbers are stored only as a keyed fingerprint plus the last 4 digits (shown as `••••9012`). Bank account numbers are encrypted. Labs, VVBs and buyers never see farmer names, contacts or GPS locations.

SIGN-IN SECURITY

Passwords have at least 12 characters with letters and digits. A temporary password must be changed at first sign-in. Accounts lock for 15 minutes after 5 failures. Sign-in is rate-limited, sessions are short-lived and refresh tokens rotate.

ORGANIZATION ISOLATION

Users only see records of their own organization. A record outside their scope returns "not found", so its existence is not revealed.

ENVIRONMENTS

Records carry an environment (LIVE / TEST / DEMO). Calculation modules not yet signed off are blocked in production.

NO SILENT RECALCULATION

A recalculation after verification or issuance creates a new run and invalidates the old package; nothing already issued is changed.

5. How to read the stage chapters

Each stage chapter (S0–S17) follows the same layout:

- **Purpose:** why the stage exists.
- **Who:** the roles involved.
- **Steps:** the actions in order, with the screen to use and the expected result.
- **Controls:** what the platform checks or enforces.
- **Output:** what the stage hands to the next one.
- **How to test:** positive checks marked **✓**, and negative tests marked **✗ must be refused**. A refusal is a pass: it proves the control works.

Screen names in **bold** match the application menu exactly, e.g. **Projects** → **MRV**. The example values are the ones used in the recorded test runs.

Tester tips

- Keep two or three browser windows open (normal, private and a second browser). Each holds its own sign-in, so testers can act as different people side by side.
- Date boxes use month/day/year.
- For field work, use the phone view (F12, then Ctrl+Shift+M in the browser).

S0 Environment and access

PURPOSE

Bring up a working environment with organizations and user accounts so that each role can be exercised.

WHO

System administrator (command line, once), then Platform Admin (in the application).

Steps

#	Who	Action	Expected result
1	System admin	Create the SQL Server database, fill in the settings file (database address, secret keys), run <code>manage.py setup</code> . This runs all database migrations and loads reference data.	About 140 tables; 96 permissions, 21 system roles, the PLATFORM organization and the "Personal data processing" consent type.
2	System admin	Run <code>manage.py bootstrap-admin</code> .	The first Platform Admin, with a temporary password.
3	System admin	Start the backend and the web application; open the site.	Sign-in page.
4	Platform Admin	Sign in, set a new password.	Forced password change, then the dashboard.
5	Platform Admin	Administration → Organizations : create Developer, Laboratory, VVB, Buyer, Registry and Farmer-group organizations.	Organizations listed and active.
6	Platform Admin	Administration → Users → New user : one user per role (Appendix A), each with a temporary password.	Users created; each must change password at first sign-in.

How to test

- ✓ Sign in as the project manager: the menu shows only project-developer pages (Field operations, Projects, MRV ...), not Administration.
- ✓ A user with a temporary password cannot use any page until the password is changed.
- ✗ Five wrong passwords lock the account for 15 minutes, and the error message does not reveal whether the email exists.
- ✗ A lab technician opening the MRV page directly by address gets "You don't have access to this page".

S1 Catalog: standards and activities

PURPOSE

Define the carbon standards the platform works with, and the project activities offered under each. Everything downstream (methodology candidates, project set-up) filters on this catalog.

WHO / WHERE

Platform Admin · **Projects** → **Standards & activities**;
Administration → **Consent types**.

#	Action	Expected result
1	Check Administration → Consent types .	"Personal data processing" (DATA_PROCESSING, v1) is Active. It is required to activate a farmer.

#	Action	Expected result
2	Add standard: Code <code>VCS</code> , Name "Verified Carbon Standard", Programme owner "Verra", Type Voluntary. Similarly <code>GS</code> (Gold Standard) and CCTS (Bureau of Energy Efficiency, compliance) as needed.	Standard listed.
3	Add activity: Code <code>IALM</code> , "Improved agricultural land management", Offered under ✓ Verified Carbon Standard. Others: <code>AGR-SOC</code> (GS), <code>ARR</code> (VCS) ...	The activity shows "Offered under: Verified Carbon Standard".

- ✓ An activity not linked to a standard never appears later in project set-up or methodology candidates.
- ✗ A non-admin user does not see **Standards & activities** in the menu.

S2 Methodology and calculation module

PURPOSE

Encode a carbon methodology (e.g. Verra VM0042 v2.2) as a versioned, approved set of rules: when it applies, what must be monitored, how samples are taken. Then attach the **calculation module** that implements its equations.

WHO / WHERE

Methodology Specialist 1 (author), Methodology Specialist 2 (approver) · **Projects** → **Methodologies**.

Steps

#	Who	Action	Expected result
1	Spec. 1	Add a methodology: Code VM0042, Name "Improved Agricultural Land Management", Standard VCS, Activity IALM.	Methodology created.
2	Spec. 1	New draft version, label 2.2. Details: Effective from 10/21/2025; Source "Verra VM0042 v2.2".	Draft version.
3	Spec. 1	Calculation tab → Calculation module: choose "VM0042 v2.2 – SOC measure & remeasure (QA2) + default-factor emissions (QA3)" → Use this module.	The draft receives 8 calculation rules, 13 monitoring rules (organic carbon and soil mass from the lab; fertiliser, fuel, lime, residue burning, irrigation per farm; risk rating per project) and a sampling rule (minimum 3 samples per stratum, core details required).
4	Spec. 1	Applicability tab: rule A1 <code>standard_code EQUALS "VCS"</code> , rule A2 <code>activity_code EQUALS "IALM"</code> , each "Not applicable" if it fails.	Two rules.
5	Spec. 1	Submit for approval.	Status In review .
6	Spec. 2	Open the version → Approve.	Status Approved ; all rules become read-only. "Calculation: Not production ready" is shown.
7	Spec. 1 → 2	(Production only, after the expert review.) Request production readiness with the expert's evidence summary; Spec. 2 approves.	Module allowed in production.

Controls

- An approved version is immutable; changes need a new version.
- A version can only be offered to projects that start on or after its effective date.
- The Calculation tab lists every data item the module needs and every interpretation of the methodology text it uses, so the expert reviewer can check each one.
- One version holds one module. A methodology with several modules (e.g. Gold Standard SOC) uses labels such as 1.0-ZT and 1.0-CC-A3.

How to test

- ✗ The author tries **Approve** on their own submission: refused.
- ✗ Editing a rule on the approved version: not possible.
- ✓ The module can be switched while the version is a draft, and the generated rules change with it.

S3 Farmer onboarding

PURPOSE

Register each participating farmer with verified identity (KYC) and recorded consent to process their personal data, before they can join a project.

WHO / WHERE

Project Manager, QA Officer · **Field operations** → **Farmers**.

#	Who	Action	Expected result
1	PM	Register farmer: managing organization, full name "Ramesh Patil", phone, village Pimpalgaon, district Nashik, state Maharashtra, country IN.	Code FRM-2026-000001, status Draft .
2	PM	Move to Registered .	Registered.
3	PM	KYC tab: upload ID document (PDF/PNG/JPG), ID type, ID number 1234 5678 9012 → submit.	KYC pending; number shown as9012.
4	QA	Same farmer → KYC → Approve .	KYC verified .
5	PM	Consents tab: "Personal data processing", text version v1, captured by "Paper signed" → Record consent .	Consent recorded; checklist item ticked.
6	PM	Activate .	Status Active .
7	PM	(Before payouts) Add the farmer's bank account; Finance verifies it.	Primary verified bank account; only last 4 digits shown.
8	PM	(Optional) Link a farmer login to the profile, for self-service.	Farmer sees "My farm" pages.

- ✗ The PM tries to approve the KYC they submitted: refused or no button.
- ✗ **Activate** without verified KYC and recorded consent: blocked; the "still needed" checklist names what is missing.
- ✓ The full ID number is never displayed again after submission.

S4 Farm registration and GIS verification

PURPOSE

Record each farm's exact boundary, ownership and land records. Check that it does not overlap other farms, and have a GIS specialist verify it. Credits are only ever claimed on verified land.

WHO / WHERE

Project Manager, GIS Specialist · **Field operations** → **Farms**.

#	Who	Action	Expected result
1	PM	New farm: farmer, farm name "Pimpalgaon plot 1", land tenure Owned, declared area 2 ha, location → Create farm and draw boundary .	FARM-2026-000001, Draft .
2	PM	Boundary tab: click corners on the map, paste coordinates, or import GeoJSON/KML → Check boundary → Save as new boundary version .	"Measured by SQL Server ≈ 2.2 ha". Boundary versions are kept.
3	PM	Documents tab: upload the land record (e.g. 7/12 extract) as category LAND_RECORD.	Document stored with checksum.

#	Who	Action	Expected result
4	PM	Ownership tab: owner type Farmer, relationship Owner, share 100 %, title/survey reference, valid from, and Evidence document (the newest land record is preselected) → save.	Ownership record showing "Evidence: 7/12 extract ...".
5	PM	Submit.	Submitted ; the overlap check runs against all other farms.
6	GIS	Ownership → Verify the record; header → Start review ; Overlaps tab → clear or resolve anything listed; Verify .	Farm Verified ; land-records card shows "Verified".

Supporting information on the farm page

- **History** tab: land use, crops and practices by year (historical and proposed), used for baseline and eligibility.
- **Map**: opens in **3D** by default (terrain view). The boundary editor stays 2D for precise drawing, and browsers without 3D support fall back to 2D automatically.
- **External data** tab: weather, soil reference and satellite indicators (Chapter 7). These are for reference only and never feed the calculation.
- **Land-records card**: shows one of Missing, Uploaded (ownership to be recorded), Awaiting GIS review, Not linked to ownership, or Verified, with a hint and buttons for the next action.
- **✗** The submitter looks for **Verify**: not available.
- **✗** Uploading a file renamed to `.pdf` that is not a PDF: rejected (the type is detected from the content).
- **✓** A second farm drawn across the first one is flagged in **Overlaps** and cannot be verified until resolved.

S5 Project and eligibility

PURPOSE

Group verified farms into a carbon project. Each farm's carbon rights must be documented. The project's standard, activity, crediting period and baseline are recorded, and it passes an eligibility review.

WHO / WHERE

Project Manager, GIS Specialist, QA Officer · **Projects** → **Projects**.

#	Who	Action	Expected result
1	PM	New project: organization, name "Nashik soil carbon pilot", country IN, region Maharashtra, planned start 11/01/2025.	PRJ-2026-000001, Draft .
2	PM	Start data collection.	Data collection.
3	PM	Farms tab: add the verified farm with its agreement/evidence reference (e.g. "Participation agreement clause 4").	Farm listed with a "Rights" chip.
4	PM	Boundary tab: view the project area.	Area computed by SQL Server from member farms.
5	PM	Team tab: add QA officer and other team members.	Team listed.
6	PM	Standard & activity tab: select VCS and IALM.	Both "(current)".
7	PM	Crediting & baseline: crediting period 11/01/2025–10/31/2035; baseline period and description "Conventional tillage, residue burning".	Both recorded.
8	PM	Submit for eligibility review.	Eligibility review.
9	GIS	Boundary tab → Accept boundary.	Review: Accepted.
10	QA	Carbon rights tab → Verify ; then Approve eligibility.	Standard selected.
11	PM	Confirm activity.	Activity selected.

- ❌ The PM looks for **Approve eligibility** on their own submission: not shown.
- ❌ Adding an unverified farm to the project: refused (FARM_NOT_VERIFIED).
- ❌ Adding a farm that conflicts with another project (overlapping participation): refused until the conflict is acknowledged with a note (CONFLICTS_REQUIRE_ACKNOWLEDGEMENT); the conflict is then reviewed during eligibility.
- ❌ Approving eligibility while a participating farm still has an OPEN overlap flag: refused (REQUIREMENTS_NOT_MET). The GIS specialist must first clear or confirm the flag on the farm's **Overlaps** tab.
- ❌ Carbon-rights shares for one farm totalling more than 100 %: refused.

S6 Methodology selection and lock

PURPOSE

Decide which approved methodology version governs the project. The rules engine proposes candidates, a specialist recommends one, and the project locks on it. From then on all monitoring follows that version.

WHO / WHERE

Project Manager, Methodology Specialist · Project → **Methodology** tab.

#	Who	Action	Expected result
1	PM	Evaluate candidates.	"VM0042 2.2: Applicable", with each rule shown (A1 ✓, A2 ✓). Status Methodology review.
2	Spec. 1	Expand the candidate → Recommend.	"Recommended by ...".
3	PM	Confirm & lock.	Lock card "VM0042 2.2"; status Methodology confirmed.

- ✗ The recommending specialist tries **Confirm & lock**: not allowed.
- ✗ Re-evaluating after the recommendation and then locking: refused with `EVALUATION_OUTDATED`; a fresh recommendation is required.
- ✓ **Status history** shows: Draft → Data collection → Eligibility review → Standard selected → Activity selected → Methodology review → Methodology confirmed.
- ✓ A project starting before the version's effective date gets no candidate for that version.

S7 MRV planning, strata and control sites

PURPOSE

Plan the monitoring the methodology requires. Divide the project land into **strata** (areas with similar soil and climate, sampled separately) and, where the methodology requires a baseline from comparable land, register **control sites**.

WHO / WHERE

MRV Manager, QA Officer, GIS Specialist · **Projects** → **MRV** → project.

#	Who	Action	Expected result
1	MRV	Create MRV plan: frequency Annual, approach Measure-and-remeasure, dates, required evidence "Field photos, GPS, lab report" → create → Submit for approval .	The methodology's monitoring rules are added automatically; Submitted .
2	QA	MRV plans → Approve .	Approved; project MRV planned .
3	MRV	Stratification: project stratum S1 , tick the project farm(s), characteristics SOIL_TEXTURE, SOIL_GROUP, PRECIPITATION_MM (e.g. 750).	Stratum draft.
4	MRV	(VM0042) Switch to Control site : create C1, C2, C3, one non-project farm each, Represents S1 , with the same texture and soil group and rainfall within 100 mm.	Control strata draft.
5	GIS	Approve each stratum.	Checklist: "3 of at least 3" control sites; "Every stratum has a control site".

Controls

- MRV is only possible on a project whose methodology is locked.
- Control sites must be verified farms of the same organization that are *not* in the project. Their sampling points are placed inside the control farm, not the project boundary.
- If the methodology leaves something unconfigured, a yellow `CONFIGURATION_REQUIRED` note appears. The approver must explicitly acknowledge it, and the acknowledgement is audited.

Open decision: strata inside a farm

Today a stratum is built from whole farms (each farm belongs to one stratum). VM0042 allows, and the rice methodology VM0051 requires, splitting a single field into several strata when soil or management differs within it. Research with methodology quotes is complete; the proposal is to allow stratum polygons drawn inside farms while keeping whole-farm strata as the default. This is awaiting a management decision (Chapter 9).

- **✗** A stratum approved by the person who created it: refused.
- **✗** A control site that matches the project stratum poorly (different soil group, or rainfall more than 100 mm apart): flagged; the calculation reports `CONTROL_SITES_REQUIRED`.

S8 Monitoring periods, sampling and field collection

PURPOSE

Define the monitoring periods and generate statistically placed sampling points. Field collectors then visit each point with a phone and record GPS, depth, core details, a photo and a checklist.

WHO / WHERE

MRV Manager, GIS Specialist, Field Supervisor, Field Collector · **Projects** → **MRV**; **Field operations** → **Field work** (phone).

#	Who	Action	Expected result
1	MRV	Monitoring periods: Period 0 , purpose BASELINE, 2025-11-01 → 2026-05-31 (the sampling at time zero), and Period 1 , purpose MONITORING, 2026-06-01 → 2027-05-31 (the re-measurement). For each: Mark planned → Start period → Open data collection .	Periods in Data collection ; project Monitoring .
2	MRV	Sampling design per period: code, type (stratified random / systematic grid), method "Soil auger", depth 0–50 cm, target precision and confidence, number of samples per stratum and per control site (e.g. 3).	Draft design.
3	GIS	Approve design .	Approved.
4	MRV	Generate sampling points .	"Points generated (SQL Server validated)"; points shown on the map inside each farm.
5	Supervisor	Points & assignments → select unassigned → choose collector → Assign .	Assigned.
6	Collector	Field work (phone) → Start collection on each point: coordinates, time, depth 0/50, probe inside diameter (e.g. 21.5 mm), number of cores (e.g. 4), checklist, photo → Save & submit .	Records Submitted .
7	Supervisor	Accept each record (or return it with a reason).	Records Accepted .

Controls

- Collectors see and change only their own assigned points and approve nothing.
- Field records are validated: no future dates, whole-number depths, and a location consistent with the planned point. Relocating a point needs an approval.
- Core details (probe diameter, number of cores) are mandatory where the methodology requires them; they are needed to compute soil mass.
- ❌ The collector tries to accept their own record: not allowed.
- ❌ Submitting a field record with a future collection date: refused (`REQUIREMENTS_NOT_MET` , "a collection time that is not in the future"). A draft may be saved incomplete; the checks run at submission.
- ✅ Point generation on a farm too small for the requested count: `INSUFFICIENT_AREA` .

S9 Laboratory analysis

PURPOSE

Analyse soil samples under an unbroken chain of custody. Samples are registered and sealed in the field, shipped, received with a seal check, analysed, and QA-approved by a second lab person.

WHO / WHERE

MRV Manager, Collector, Supervisor (developer side) · Lab Manager, Lab Technician, Lab QA (lab side) · **MRV** → **Samples & laboratory; Laboratory.**

#	Who	Action	Expected result
1	MRV	Propose a lab engagement : laboratory "Soil Test Lab", tick the parameters (VM0042: organic carbon V42_OC , soil mass V42_SOIL_MASS).	Proposed.
2	Lab manager	Laboratory → Engagements → Accept.	Active.
3	Collector	On each accepted field record register two depth samples (0–30 cm and 30–50 cm), e.g. "Topsoil core", 500 g, label BAG-01, then seal (SEAL-0001).	Samples SMP-2026-... Sealed ; one lab test per parameter created automatically.
4	Supervisor	Create shipment to the lab, add samples, Dispatch.	Dispatched.
5	Lab tech	Incoming : for each sample record the seal observed and condition → Record receipt ; Samples : accession number → Register.	Received; Registered.
6	Lab tech	Worklist → each test → Start test → method (e.g. Walkley-Black), value (organic carbon e.g. 12.5 g/kg; dry fine-earth mass e.g. 280 g), unit, time → Save result → upload the PDF lab report → Submit for QA.	Submitted.
7	Lab QA	Laboratory → QA → Review → Start QA review (12 automatic checks) → decision APPROVED with notes.	Results Approved ; samples Analysed.
8	MRV	Lineage on a result.	Result → test → sample → field record → point → farm → project.

Controls

- The engagement is two-sided: the proposer cannot accept it.
- The lab sees only the samples shipped to it, through an allow-listed interface: no farmer, farm or GPS data.
- A seal mismatch at receipt fails the lab QA custody check; lab QA cannot approve a result with a failed check (QA_CHECKS_FAILED) and must reject it.
- A result cannot be submitted without its PDF report (REPORT_REQUIRED). Approved results are immutable; corrections create a new version.
- Custody events and QA reviews are append-only.
- ✗ The lab technician tries to approve their own result: refused.
- ✗ Recording a receipt with a seal number different from the one applied in the field: lab QA custody check fails ("seal number observed at receipt differs from the seal applied in the field").
- ✗ Submitting a result without the PDF report: refused (REPORT_REQUIRED).
- ✗ Lab QA approving a result whose automatic checks failed: refused (QA_CHECKS_FAILED).
- ✗ The lab technician opens the MRV page: "You don't have access to this page".

S10 Monitoring data and dataset QA

PURPOSE

Record the non-laboratory data the methodology needs: farm activity (fertiliser, diesel, residue burning, irrigation), project-level factors (e.g. risk rating), and per-stratum or per-tree values. Then freeze the period's data into a QA-approved dataset.

WHO / WHERE

MRV Manager, QA Officer · **MRV** → **Monitoring data; Datasets & QA.**

#	Who	Action	Expected result
1	MRV	Monitoring data , per farm, phase BASELINE : one record per historical year (2023, 2024, 2025), e.g. synthetic N fertiliser kg, diesel litres, residue burnt kg.	Records listed.
2	MRV	Per farm, phase PROJECT , observed inside Period 1: the same activities under the new practice.	Recorded.
3	MRV	Project level: e.g. <code>V42_NPR</code> = 12 (% non-permanence risk). Other methodologies record at Stratum or Sampling point level (e.g. trees alive, biomass).	Recorded.
4	MRV	Datasets & QA : choose the period → Create dataset version → Submit for QA .	Submitted: a frozen, SHA-256 hashed snapshot of every record, sample result and stratum.
5	QA	Start QA review (15 automatic checks) → QA result PASS → Approve dataset .	Dataset and period Approved .

- ✓ The QA review warns "analysis pending" only for points whose lab results are not yet approved.
- ✗ The MRV manager approving their own dataset: refused.
- ✓ Changing a record after the dataset was submitted does not change the frozen snapshot; a new dataset version is needed.

Known issue (workaround in place)

After a dataset has been submitted, do not press **Open data collection** on that period again; it can leave the period stuck. Testers should avoid this until it is fixed.

S11 Carbon calculation and internal QA

PURPOSE

Apply the locked methodology's equations to the approved data and produce the credit quantity. Every step is traceable to an equation and to the frozen inputs.

WHO / WHERE

Calculation Analyst, QA Officer · **Projects** → **Calculations.**

#	Who	Action	Expected result
1	Analyst	Choose project and Period 1 ; review the readiness panel.	No blockers; only the warning <code>NOT_PRODUCTION_READY: non-production use only</code> .
2	Analyst	New calculation run → Check readiness & freeze inputs .	Inputs frozen with SHA-256.

#	Who	Action	Expected result
3	Analyst	Execute.	Step-by-step outputs, each citing its equation. For VM0042: reference soil mass; SOC stocks per stratum and campaign; emissions per source; SOC change; uncertainty; buffer; VCU_TOTAL (whole tonnes, rounded down).
4	QA	Calculation QA: Findings if needed (analyst responds, QA resolves) → Approve.	Run Approved ; outputs re-verified against their hashes.
5	Analyst / QA	Generate the calculation report (PDF) and internal verification readiness.	Report with content and PDF hashes; readiness READY with a frozen package manifest.

Typical blockers (the readiness panel explains each one)

Code	Meaning
PREVIOUS_PERIOD_REQUIRED	The baseline (Period 0) dataset is not approved.
CONTROL_SITES_REQUIRED	Fewer than 3 sampled control sites (VM0042).
INSUFFICIENT_SAMPLES	Too few samples in a stratum and campaign.
DEPTH_INCREMENTS_REQUIRED / SAMPLING_DEPTH_INSUFFICIENT	Only one depth layer, or sampled too shallow (e.g. zero tillage needs 50 cm).
MISSING_REQUIRED_INPUT	A required data item is missing, e.g. core details or an emission factor.
MODULE_NOT_SELECTED	The methodology version has no calculation module.
DENSITY_EXCEEDED	VM0047 census approach used with more than 50 trees per hectare.

- ❌ The analyst approving their own run: refused.
- ❌ In the production environment, a module without production-readiness sign-off: blocked.
- ✅ Any change to an input after the freeze is detected before execution, and the run refuses to execute.
- ✅ An open blocking finding invalidates a READY readiness immediately.

S12 Independent verification (VVB)

PURPOSE

Have an accredited independent body review the evidence package and state whether the reductions are verified, and in what quantity.

WHO / WHERE

Project Manager, VVB Reviewer · Project → **Verification** tab; **Verification** → **VVB workspace**.

#	Who	Action	Expected result
1	PM	Propose a VVB assignment for the period: VVB organization "Verify Co".	Assignment Proposed .
2	VVB	VVB workspace → Accept with a conflict-of-interest declaration.	Accepted ; project status Verification .
3	PM	Submit package (requires READY internal readiness).	Submission created from the frozen manifest.
4	VVB	Open the package: calculation report, outputs, evidence documents. Raise findings or corrective action requests if needed; the project responds.	Allow-listed view: no farmer names, contacts or bank data.
5	VVB (2nd reviewer if findings were raised)	Decision : outcome VERIFIED (or not), rationale, verified quantity (e.g. 24 tCO ₂ e), and the verification report PDF.	Decision code (e.g. VDEC-2026-000002); project Verified .

Controls

- The VVB interface is separate and scoped to the accepted assignment. Anything else returns "not found".
- A decision is immutable and tied by hash to the exact package it reviewed.
- If the project recalculates, the old submission is superseded. A new package and a new decision are required, as happened in the recorded test run (Chapter 8).
- The VVB raises and closes findings; the project can only respond. A VVB reviewer who raised findings on a package cannot record its decision: a second reviewer of the same VVB must (`SEPARATION_OF_DUTIES`).
- ✗ Submitting a package before the calculation is approved and readiness is READY: refused (`NO_READY_PACKAGE`).
- ✗ The VVB reviewer who raised findings records the decision: refused.
- ✗ A VVB user from another VVB organization opening the package: not found.

S13 Registry submission and issuance

PURPOSE

Submit the verified period to the carbon registry and record the registry's acceptance and the issuance: quantity, vintage and serial number ranges, backed by the registry's own documents.

WHO / WHERE

Registry Manager, QA Officer · **Registry** → **Registry**.

#	Who	Action	Expected result
1	Registry mgr	Record the registry account and the project registration (registry project ID plus registry response PDF).	Account Active; registration Registered.

#	Who	Action	Expected result
2	Registry mgr	Create an issuance submission for the verified period; attach required documents (e.g. PDD).	Draft; eligibility re-checked (current VVB decision, approved run).
3	Registry mgr	Freeze the submission.	Snapshot frozen with SHA-256.
4	Registry mgr	Record submitted (registry reference plus receipt PDF), then Record response : ACCEPTED (acknowledgement PDF).	Submitted → Accepted.
5	Registry mgr	Record issuance : registry issuance ID, date, quantity 24 VCU, vintage 2026, serial range (e.g. ...-0001 to ...-0024), issuance statement PDF.	Issuance Recorded .
6	QA	Confirm the issuance against the registry statement.	Issuance Confirmed ; batch Issued ; project Issued .

- ❌ The registry manager confirming their own issuance: refused (HTTP 403).
- ❌ Recording (or confirming) issuances that together exceed the VVB-stated verified quantity: refused (QUANTITY_EXCEEDS_VERIFIED).
- ✅ Registry references and the frozen snapshot cannot be changed afterwards (database triggers).

S14 Credit ledger

PURPOSE

Track ownership of every issued credit in an append-only ledger. Credits can be reserved, transferred and retired, and can never be double-spent.

WHO / WHERE

Credit Manager, QA Officer · **Registry** → **Issued credits**; **Credit ledger**; buyers: **Credits** → **My credits**.

#	Who	Action	Expected result
1	Credit mgr	Issued credits → the issued batch → Open in ledger .	Opening requested.
2	QA	Confirm the opening.	24 VCU Available in the developer's inventory, by serial range.

Controls

- Ledger entries are append-only, and database triggers enforce conservation: credits in equal credits out.
- A position can be consumed only once. A retired credit is final: no un-retirement and no reversal.
- Concurrent operations on the same credits use row locks; the loser of a race is refused cleanly.
- ✗ The credit manager confirming their own opening: refused (HTTP 403).
- ✓ Inventory balances always add up to the issued quantity (available + reserved + transferred + retired).

S15 Marketplace, delivery and retirement

PURPOSE

Sell credits to KYC-verified buyers at a fixed price, record payment, deliver the credits into the buyer's holding and support retirement against a registry certificate.

WHO / WHERE

Buyer, Marketplace Compliance, Credit Manager, Finance, QA (custodian) · **Marketplace** menu.

#	Who	Action	Expected result
1	Buyer	Buyer profile : legal name, country, upload KYC document → Submit KYC .	KYC submitted.
2	Compliance	KYC review → Verify .	KYC verified .
3	Credit mgr	Listings → new listing: batch and serial range, 20 VCU at INR 850, payment window 24 h → Submit .	Listing pending approval.
4	Finance	Approve the listing.	Listing Active ; visible on the marketplace.
5	Buyer	Order 10 VCU.	Order Placed , total INR 8,500; credits reserved.
6	Buyer	Upload bank-transfer proof and record the payment (reference, exact amount).	Payment Pending confirmation .
7	Finance	Payments → Confirm payment received.	Order Paid → Transfer pending .
8	QA (custodian)	Complete the delivery transfer.	Order Completed ; buyer holds 10 VCU (My credits); order confirmation PDF and lineage available.

#	Who	Action	Expected result
9	Buyer	Request retirement of 4 VCU: beneficiary and reason (e.g. "Voluntary claim FY2026").	Retirement Requested .
10	QA (custodian)	Record the registry retirement reference, date and certificate PDF → Retire .	4 VCU Retired (final); buyer holds 6 available.

- ❌ The buyer confirming their own payment: refused (HTTP 403).
- ❌ A buyer without verified KYC placing an order: refused.
- ✅ The public listing discloses no farmer, farm, location, KYC, lab or calculation-internal data.
- ✅ Refunds (before delivery) release the reservation; credits never leave the seller.

S16 Revenue sharing, settlement and farmer payouts

PURPOSE

Turn sales revenue into farmer payments under an approved benefit-sharing agreement, with a reproducible settlement calculation, bank-verified payouts and reconciliation against the bank statement.

WHO / WHERE

Project Manager, Finance Managers 1–4 · **Finance** → **Revenue & costs, Revenue sharing, Settlements, Payouts.**

#	Who	Action	Expected result
1	Finance	Revenue & costs: check the revenue recognised from the sale.	INR 8,500 for the project and period.
2	PM → Fin.	Revenue sharing: farmer share 60 %, deduct approved costs, rounding Half-up, effective date, source "Benefit-sharing agreement clause 4" → submit; finance approves.	Sharing version Approved .
3	PM → Fin.	Allocation across project farms (e.g. 100 % to the one farm, basis "by area") → submit; finance approves.	Allocation Approved .
4	PM → Fin.	Record a project cost : "Soil sampling and laboratory analysis", INR 1,500, with invoice PDF; finance approves.	Cost Approved .
5	Finance 2	Settlements → create → Calculate → Verify → Submit .	Gross 8,500 – costs 1,500 = distributable 7,000 ; farmers 60 % = 4,200 ; developer 2,800. "Independently reproducible: yes".
6	Finance 3	Approve the settlement.	Settlement Approved .
7	Finance 2	Create payouts from the settlement → submit.	Payout INR 4,200 to the farmer, Pending approval .
8	PM → Fin.	Add the farmer's bank account; finance verifies it (e.g. penny-drop).	Primary verified account.
9	Finance 3	Approve the payout.	Approved (bank account referenced by last 4 digits).
10	Finance 4	Initiate the payment, then Confirm paid with the bank transfer reference (NEFT) and evidence PDF.	Payout Paid .
11	Finance	Reconcile against the bank statement (reference, amount, date, statement PDF).	Result MATCHED ; payout Reconciled ; settlement Completed .

- ✗ The finance manager who calculated the settlement tries to approve it: refused.
- ✗ Approving a payout when the farmer has no verified bank account: refused with `BANK_ACCOUNT_NOT_VERIFIED`.
- ✗ The approver executing the same payout: refused (executor ≠ approver ≠ calculator).
- ✓ A failed payout is never edited; it is replaced by a new payout linked to the failed one.
- ✓ If the bank account changes between approval and execution, the payout goes **On hold**.

S17 Farmer self-service

PURPOSE

Give farmers transparent access to their own data: profile, farms, project participation and payments.

WHO / WHERE

Farmer login linked to the farmer profile · **My farm** menu: My farmer profile, My farms, My projects, My payouts.

- ✓ **My payouts** shows the INR 4,200 payout with status Reconciled.
- ✗ A farmer cannot see other farmers, the project-wide view, or any internal page (MRV, calculations, finance).

6. Supported methodologies

The platform ships **13 calculation modules**, one for each methodology option below. Each module encodes the official methodology's equations step by step and declares the data it needs. When a methodology is set up, the code and version label must be entered exactly as below so that the module can be attached.

Methodology	Code / label	Scope of the module	Key data recorded	Credit output
Verra VM0042 v2.2 Improved Agricultural Land Management	VM0042 / 2.2	Soil organic carbon by measure-and-remeasure (Quantification Approach 2) with control sites; emissions by default factors (Approach 3).	Lab organic carbon and soil mass at two depths; farm fertiliser, fuel, lime, burning; risk rating	VCU_TOTAL
Gold Standard SOC Framework v1.0 (402)	GS402 / 1.0[-suffix]	Five modules: Zero tillage (Approach 1, measured); Improved tillage (Approach 1 measured, or Approach 3 with IPCC factors); Cover crops (Approach 1, or Approach 3 with national factors).	Lab samples (A1) or SOC reference and stock-change factors (A3); farm activities; yields	GS_VER_TOTAL
Verra VM0047 v1.1 Afforestation, reforestation, revegetation	VM0047 / 1.1	Census-based approach: trees planted on farmland, at most 50 trees/ha.	Tree census; per sampled tree: alive, biomass, burned; root-shoot ratio	VCU_TOTAL
Verra VM0044 v1.2 Biochar	VM0044 / 1.2	Biochar production and application in soil and non-soil uses.	Batches produced/applied, carbon content, H/C ratio, temperature, technology	VCU_TOTAL
India CCTS FR05.002 Afforestation/reforestation	BM-FR05.002 / 1.0	CDM A/R-based tool chain for India's Carbon Credit Trading Scheme.	Plot biomass, plot area, baseline stocks, shrubs, leakage	LCCC
India CCTS FR05.001 Mangroves	BM-FR05.001 / 1.0	Mangrove afforestation/reforestation.	As above, plus planting year per stratum	LCCC
Verra VM0051 v1.1 Rice cultivation	VM0051 / 1.1	Methane from improved rice water management (Quantification Approach 3).	Per farm and season: area, days, water regime, straw, N; ≥ 3 baseline years; region	CREDITS
India CCTS AG04.002 Rice	BM-AG04.002 / 1.0	Rice methane under CCTS.	As VM0051	CREDITS
Gold Standard A/R v2.1 (403)	GS403 / 2.1	Afforestation/reforestation with volume-based biomass.	Plot volume, plot area, baseline, leakage, burning, prior issuance	GSVER_PROJECT

Gold Standard Paris alignment

Under Gold Standard rule GS 119, vintages from 1 January 2026 can only be credited under a Paris-aligned methodology. Both Gold Standard modules enforce this. Without a project-level record that Gold Standard has approved the transition (GS_PAA / GA_PAA = YES), credits for 2026+ vintages are set to zero. In production, YES must only be recorded after Gold Standard approval.

7. External data, land records and maps

7.1 External data sources

The farm page's **External data** tab shows independent reference data for each farm. It helps the GIS specialist and project team plausibility-check what farmers declare. It is **reference only**: it is never used as a calculation input, because the methodologies require measured or documented data.

Source	What it provides	How it is used	Cost / key
Open-Meteo (historical weather)	Rainfall and temperature history at the farm location.	Checks the rainfall characteristic of strata (control sites must be within 100 mm of the project stratum) and supports climate-zone selection for default factors.	Free, no key
ISRIC SoilGrids	Modelled soil properties (organic carbon, texture, bulk density, pH) for the farm location.	Indicates soil texture and group for stratification and helps plan sampling. It is <i>not</i> a substitute for laboratory analysis.	Free, no key
Microsoft Planetary Computer (Sentinel-2, Landsat)	Per-farm vegetation index (NDVI), moisture index (NDMI), cloud cover over the plot, land surface temperature.	Satellite evidence of cropping and cover-crop activity over time, to cross-check declared practices.	Free, no key

Satellite scenes with more than 10 % cloud over the plot are excluded, and at most 6 recent scenes are shown. Automated tests never call these services.

7.2 Land records

Land records (e.g. the 7/12 extract in Maharashtra, title deeds, lease agreements) are uploaded as farm documents and linked as **evidence** to the ownership record. The GIS specialist verifies the ownership against the document. The land-records card on the farm page shows which step is next:

Card status	Meaning and next action
Missing	No land record uploaded. Upload one (category LAND_RECORD).
Uploaded: record the ownership	Document present, no ownership record yet. Record ownership and pick the document as evidence.
Awaiting GIS review	Ownership recorded with evidence; GIS specialist to verify.
Not linked to ownership	The farm is verified but its ownership has no document attached. Link the evidence.
Verified	Ownership verified against the land record.

7.3 Maps

- Farm, project, stratum and sampling-point maps open in **3D** (terrain) by default; users can switch to 2D.
- The boundary editor opens in 2D for precise drawing; devices without 3D support fall back to 2D automatically.
- All areas, overlaps and point positions are computed by SQL Server's geography engine, not in the browser.

8. Test evidence and results

8.1 Automated test suites

Suite	Scope	Result
Backend (pytest, separate test database)	Business rules, separation of duties, security (tokens, lockout, cross-organization access, file spoofing), append-only triggers, concurrency (no double-spend), every calculation module including methodology worked examples.	552 passed
Frontend (Vitest)	Screens and workflows: farm, ownership evidence, external data, maps (3D default and fallback), MRV records, calculations, marketplace, finance.	153 passed; production build succeeds

8.2 Full lifecycle run: VM0042 soil carbon (project PRJ-2026-000004)

Performed through the platform's own API, with 24 separate test accounts, in the order of the stage chapters.

Stage	What happened	Outcome
Onboarding (S1–S6)	Catalog, VM0042 v2.2 with module approved by a second specialist; farmer KYC and consent; farms verified; project eligible; methodology recommended and locked.	Pass
MRV (S7–S10)	Project stratum plus 3 control sites; baseline and monitoring periods; sampling points generated; field collection with core details; 2 depth samples per record, sealed, shipped, received, analysed, lab-QA approved; farm activity data; both datasets approved.	Pass
Calculation (S11)	Readiness clean (only the "not production ready" warning); inputs frozen; executed; QA approved.	24 VCU
Verification (S12)	VVB accepted with COI declaration. After a recalculation the first package was superseded, a new package was submitted and the VVB issued decision VDEC-2026-000002 : VERIFIED, 24 tCO ₂ e.	Verified 24 tCO ₂ e
Registry (S13)	Submission frozen, submitted, accepted; issuance of 24 VCU (vintage 2026, serial range) recorded and confirmed by QA. The recorder's own confirmation was refused.	24 VCU issued
Ledger (S14)	Batch opened and confirmed by a second person; 24 VCU available.	Pass
Marketplace (S15)	Buyer KYC verified; 20 VCU listed at INR 850; 10 VCU ordered (INR 8,500); payment confirmed by seller finance (the buyer's own confirmation was refused); delivered; 4 VCU retired with certificate.	Sold 10, retired 4
Finance (S16)	60 % sharing, allocation, cost INR 1,500; settlement 8,500 – 1,500 = 7,000 distributable, reproducible; farmer payout INR 4,200 (approval refused until the bank account was verified); paid by NEFT; reconciled MATCHED.	INR 4,200 reconciled
Farmer view (S17)	Farmer login sees the payout under My payouts .	Pass

8.3 Every methodology option, calculated end to end

For each option, a methodology version was created and approved with its module, a project was locked on it, the data was recorded and approved, and the calculation was executed.

Methodology option	Credits calculated
VM0042 v2.2: measure & remeasure + default-factor emissions	24 VCU
GS402 Zero tillage, Approach 1 (measured)	16 GS VER
GS402 Improved tillage, Approach 1 (measured)	11 GS VER
GS402 Cover crops, Approach 1 (measured)	11 GS VER
GS402 Improved tillage, Approach 3 (IPCC factors)	0 *
GS402 Cover crops, Approach 3 (national factors)	0 *
VM0047 v1.1 census-based (trees on farms)	6 VCU
VM0044 v1.2 Biochar	249 VCU
CCTS FR05.002 Afforestation / reforestation	270 LCCC
CCTS FR05.001 Mangroves	280 LCCC
VM0051 v1.1 Rice (Approach 3)	3 credits
CCTS AG04.002 Rice	3 credits
GS403 v2.1 Afforestation / reforestation	127 GS VER

* Correct and expected: the test entered stock-change factors without their $\pm$ uncertainty, so the methodology's default uncertainty applies, and on a small test plot the uncertainty deduction removes the whole gain. With the $\pm$ values from the IPCC tables, the same module produces credits. The test figures come from small synthetic plots and are not forecasts of real project yields.

8.4 Defects found and fixed during testing

Issue found	Fix
Control-site sampling points could not be placed (clipped to the project boundary).	Control strata are now sampled inside their own farms; regression test added.
Calculation input freeze failed for per-stratum data.	Database migration 0023 allows stratum-level inputs.
Area-based modules missed stratum areas.	The dataset snapshot now includes all approved strata.
Dataset QA warned "analysis pending" even after results were approved.	The warning counts only points still awaiting analysis.
Only one Gold Standard SOC option could be set up per methodology.	Version labels such as 1.0-ZT allow all five side by side.
Ownership form had no way to attach the land record; land-records card misleading for verified farms.	Evidence picker added; card shows "Not linked to ownership" with guidance.
Maps always opened in 2D.	3D by default, with automatic 2D fallback.
Project status moved to Verification only at package submission, not when the VVB accepted the assignment (found by the guide walkthrough, 8.5).	Status is now updated at acceptance; regression test added.

8.5 Guide walkthrough re-test (5 October 2026)

After this guide was written, the complete procedure in chapters S0–S17 was run again from scratch, exactly as written, on a new farmer, five new farms and a new VM0042 project. Every "must be refused" test in Appendix B was included. The run used the same role accounts, plus new buyer, farmer and second VVB-reviewer accounts created through **Administration** as described in S0.

263

steps and checks executed

0

failures in the final run

48 / 48refusal tests correctly
refused**24 VCU**calculated, verified, issued;
INR 4,200 paid to the farmer

What the walkthrough found and how it was handled:

- **One defect, fixed:** the project status moved to Verification only when the package was submitted, instead of when the VVB accepted. Corrected, with a regression test.
- **Four controls stricter than first described, now documented above:**
 - eligibility cannot be approved while a farm has an open overlap flag;
 - a lab result cannot be submitted without its PDF report;
 - an issuance above the verified quantity is refused when recorded, not only when confirmed;
 - the VVB reviewer who raised findings cannot record the decision.
- **One correction to this guide:** the platform has 13 calculation modules (one per methodology option), not 14.

9. Known limitations and open decisions

9.1 Before real credits can be issued

Item	Detail	Owner
Expert sign-off of calculation modules	All 13 modules are "Not production ready". A qualified methodology expert must check each against the official document, including every listed interpretation, and the sign-off is recorded in the platform (request and second approval). Until then calculations are blocked in production.	Methodology team + external expert
Gold Standard Paris alignment	2026+ vintages under GS402/GS403 are zero unless Gold Standard approves the transition.	Management / Gold Standard
Within-farm strata	Decide whether to support strata drawn inside farms (needed for heterogeneous fields; required by VM0051). Research is complete.	Management decision pending
VM0047 v1.1 details	Equations specific to v1.1 should be re-checked against the official PDF.	Methodology team

9.2 Production infrastructure

- **Malware scanning:** the scanning hook exists; a real antivirus engine must be connected (development detects only the EICAR test file).
- **Multi-factor sign-in:** data model ready; enrolment flow not yet built.
- **Rate limiting:** in-memory today; a shared store (Redis) is needed before running more than one server.
- **Payments and registries:** buyer payments, farmer payouts and registry submissions are recorded manually with references and PDF evidence. Live bank and registry integrations are not connected.
- **Encryption-key rotation:** rotating the data encryption key needs a re-encryption job (not built). The key must be kept in a secret store with a backup.
- **Known bug:** re-opening data collection on a period whose dataset was already submitted can leave the period stuck (S10).

9.3 Not in scope of the current build

- VM0047 area-based approach; rice Quantification Approaches 1 and 2.
- Use of external satellite or soil data as calculation inputs (reference only, by design).

Appendix A: test accounts

The recorded test runs used dedicated accounts named `e2e.<role>@cc.example.com` in the local test environment. Passwords are held by the test team and are not reproduced here.

Account	Organization	Role
e2e.admin	PLATFORM	Platform Admin
e2e.meth1 / e2e.meth2	Platform staff	Methodology Specialist (author / approver)
e2e.comp	Platform staff	Marketplace Compliance
e2e.pm	DEV	Project Manager
e2e.qa	DEV	QA Officer
e2e.gis	DEV	GIS Specialist
e2e.mrv	DEV	MRV Manager
e2e.sup / e2e.col	DEV	Field Supervisor / Field Agent
e2e.analyst	DEV	Calculation Analyst
e2e.registry / e2e.credit	DEV	Registry Manager / Credit Manager
e2e.fin, fin2, fin3, fin4	DEV	Finance Manager (approver, calculator, second approver, executor)
e2e.labmgr / e2e.labqa / e2e.labtech	LAB	Lab Manager / Lab Manager (QA) / Lab Technician
e2e.vvb (+ a second reviewer)	VVB	VVB Reviewer
e2e.buyer	BUY	Buyer
e2e.farmer	E2EFPO	Farmer (linked to farmer profile "Asha Patil")

Appendix B: negative-test checklist

Each line must be **refused** by the platform. Tick when confirmed.

☐	Stage	Attempt	Expected refusal
☐	S0	Lab technician opens the MRV page	No access
☐	S0	Five wrong passwords	Account locked 15 min
☐	S2	Methodology author approves own version	Separation of duties
☐	S3	PM approves own KYC submission	Separation of duties
☐	S3	Activate farmer without KYC/consent	Requirements not met
☐	S4	Farm submitter verifies own farm	No Verify action
☐	S4	Upload a fake PDF (wrong content type)	File rejected
☐	S5	PM approves own eligibility submission	Not shown / refused
☐	S5	Add a farm with an open boundary conflict, no acknowledgement	CONFLICTS_REQUIRE_ACKNOWLEDGEMENT

☐	Stage	Attempt	Expected refusal
☐	S5	Approve eligibility with an OPEN overlap flag	REQUIREMENTS_NOT_MET
☐	S6	Recommending specialist locks the methodology	Not allowed
☐	S6	Lock after re-evaluation	EVALUATION_OUTDATED
☐	S7	Stratum creator approves own stratum	Separation of duties
☐	S8	Collector accepts own field record	Not allowed
☐	S9	Lab technician approves own result	Refused
☐	S9	Submit a lab result without the PDF report	REPORT_REQUIRED
☐	S9	Receipt with wrong seal number, then lab QA approves	Custody FAIL; QA_CHECKS_FAILED
☐	S10	MRV manager approves own dataset	Separation of duties
☐	S11	Analyst approves own calculation	Separation of duties
☐	S12	Submit package before readiness is READY	NO_READY_PACKAGE
☐	S12	VVB reviewer who raised findings records the decision	Separation of duties
☐	S13	Registry manager confirms own issuance	HTTP 403
☐	S13	Record an issuance above the verified quantity	QUANTITY_EXCEEDS_VERIFIED
☐	S14	Credit manager confirms own ledger opening	HTTP 403
☐	S15	Buyer confirms own payment	HTTP 403
☐	S15	Unverified buyer places an order	Refused
☐	S16	Settlement calculator approves own settlement	Separation of duties
☐	S16	Payout approval without verified bank account	BANK_ACCOUNT_NOT_VERIFIED
☐	S16	PM verifies the bank account they added	Refused
☐	S16	Payout approver executes the payout	Separation of duties
☐	S17	Farmer opens another farmer's data	Not found

Appendix C: troubleshooting

What you see	What to do
"Too many sign-in attempts"	For testing, raise the sign-in rate limit in the server settings and restart the backend.
A button is missing	Wrong user for that step (see the role table), or it is your own submission (separation of duties).
<code>REQUIREMENTS_NOT_MET</code>	Read the "still needed" checklist on the page.
No methodology candidate	The version is not Approved, its effective-from date is after the project start, or the standard/activity does not match.
<code>EVALUATION_OUTDATED</code>	Candidates were re-evaluated after the recommendation; the specialist must recommend again.
"Invalid data" on field collection	The message names the field: usually GPS accuracy (leave empty), depth (whole numbers) or a future date.
<code>INSUFFICIENT_AREA</code> on point generation	Reduce the sample count or use a larger boundary.
Seal mismatch at lab receipt	The seal observed must equal the seal applied in the field.
<code>REPORT_REQUIRED</code> when submitting a lab result	Attach the PDF lab report to the result, then submit again.
Calculation blocker	See the blocker table in S11; the readiness panel names the stratum or data item.
Gold Standard credits are zero for 2026+	Paris-alignment gate: record $GS_PAA / GA_PAA = YES$ only after Gold Standard approval.
Approach 3 credits are zero	Stock-change factors entered without $\pm$ uncertainty; record the $\pm$ values from the source tables.

Appendix D: glossary

Term	Meaning
dMRV	Digital measurement, reporting and verification.
Methodology	The official rulebook (e.g. Verra VM0042) that defines when a project qualifies and how its credits are calculated.
Calculation module	The platform's implementation of a methodology's equations, attached to a methodology version.
SOC	Soil organic carbon.
Stratum	An area with similar conditions (soil, climate, management) that is sampled and calculated as one unit.
Control site	Comparable land outside the project, used under VM0042 to measure what would have happened without the project (the baseline).
Monitoring period	The time window for which data is collected and credits are calculated; Period 0 is the baseline sampling.

Term	Meaning
Dataset	The frozen, QA-approved collection of all monitoring data for a period.
Uncertainty deduction	Credits removed when measurement uncertainty is above the methodology's threshold.
Buffer	Share of credits set aside in a pooled reserve against future reversal (non-permanence risk).
VVB / ACVA	Validation and Verification Body (Verra, Gold Standard) / Accredited Carbon Verification Agency (India CCTS).
VCU / GS VER / CCC	Credit units: Verified Carbon Unit (Verra), Gold Standard Verified Emission Reduction, Carbon Credit Certificate (India CCTS).
Vintage	The year in which the reduction or removal happened.
Retirement	Permanent use of a credit to claim an offset; it can never be resold.
Separation of duties	The rule that the person who submits something cannot approve it.
SHA-256 hash	A digital fingerprint; any change to the data changes the fingerprint, so tampering is detectable.
KYC	Know Your Customer: identity verification of farmers and buyers.

End of document.